Exam Result Prediction Using Machine Learning

Project Report. Submitted by: [Your Name], [Roll No. / Course / Institution], [Date]

1. Abstract

This project builds a machine learning model that predicts whether a secondary-school student will pass or fail the final exam. Using the UCI Student Performance dataset (Mathematics course, 395 students), the final grade (G3) was converted into a binary Pass/Fail label, and a Logistic Regression classifier was trained on five features: weekly study time, past failures, absences, and the first- and second-period grades (G1, G2). On a held-out test set of 79 students the model reached an accuracy of 88.6%, precision of 0.94, recall of 0.89 and an F1-score of 0.91 for the Pass class. The results show that earlier grades and study-related factors are strong indicators of final exam outcome, and that a simple, interpretable model can support early identification of students at risk of failing.

2. Introduction

Identifying students who are likely to fail before the final exam gives teachers time to offer extra support. Manual identification is subjective and slow, whereas a data-driven model can screen an entire class consistently. This project frames the problem as a **binary classification task**: given a student's academic and behavioural record, predict whether the final result will be Pass (1) or Fail (0).

2.1 Objectives

- Prepare and explore the student performance dataset.
- Define a Pass/Fail target from the final grade.
- Train a classification model to predict the result.
- Evaluate the model with standard metrics and visualisations.
- Demonstrate prediction for a new, unseen student.

2.2 Tools and Technologies

- **Language and environment:** Python 3 in Google Colab

- **Libraries:** pandas, NumPy, Matplotlib, scikit-learn
- **Algorithm:** Logistic Regression

3. Dataset Description

The data comes from the UCI Machine Learning Repository's *Student Performance* dataset, supplied as a ZIP archive containing separate files for the Mathematics course (`student-mat.csv`) and the Portuguese course (`student-por.csv`). This project uses the **Mathematics file**.

Property	Value
Records (students)	395
Original columns	33 (34 after adding the Result column)
Numeric columns	17 (including the added Result)
Categorical columns	17
Missing values	None
Duplicate rows	None

The attributes cover demographics (sex, age, address, family size), family background (parents' education and jobs), school-related factors (study time, past failures, extra support, absences) and lifestyle factors (alcohol use, going out, health). The three grade columns are G1 (first period), G2 (second period) and G3 (final grade), each on a 0 to 20 scale.

4. Methodology

4.1 Data Loading

The ZIP archive was uploaded to Colab and extracted. It contained a nested ZIP, which was extracted in a second step to reach the CSV files. The Mathematics file was read with a semicolon separator.

4.2 Target Creation

The final grade G3 was converted to a binary label:

- **Pass (1):** G3 is 10 or above
- **Fail (0):** G3 is below 10

This gives 265 Pass and 130 Fail students, so roughly 67% of the data is the Pass class. The classes are moderately imbalanced.

4.3 Data Cleaning

The dataset was checked for missing values and duplicate rows. None were found, so the dataset remained at 395 rows after the duplicate-removal step.

4.4 Feature Selection

Five numeric input features were selected:

Feature	Meaning	Mean	Min	Max
studytime	Weekly study time (1 to 4 scale)	2.04	1	4
failures	Number of past class failures	0.33	0	3
absences	Number of school absences	5.71	0	75
G1	First-period grade	10.91	3	19
G2	Second-period grade	10.71	0	19

The statistics show that absences are right-skewed (most students have few, a small number have very many, with a maximum of 75), and that G2 includes students scoring 0.

4.5 Exploratory Analysis

Two visualisations were produced. A bar chart of Pass versus Fail counts confirms the class imbalance. A bar chart of average pass rate by study-time level examines whether students who study more tend to pass more often.

4.6 Train/Test Split and Scaling

The data was split 80% training (316 students) and 20% testing (79 students), using a fixed random seed (42) and **stratified sampling** so that both sets keep the same Pass/Fail proportion. Features were standardised with `StandardScaler`, which was **fitted on the**

training data only and then applied to the test data. This avoids leaking test-set information into training.

4.7 Model Training

A Logistic Regression model (scikit-learn defaults, random seed 42) was trained on the scaled training data. Logistic Regression was chosen because it is well suited to binary outcomes, fast to train on small data, and easy to interpret.

5. Results

5.1 Performance Metrics

The model was evaluated on the 79 unseen test students.

Metric	Score
Accuracy	0.89 (88.6%)
Precision (Pass)	0.94
Recall (Pass)	0.89
F1-Score (Pass)	0.91

5.2 Classification Report

Class	Precision	Recall	F1-Score	Support
Fail	0.79	0.88	0.84	26
Pass	0.94	0.89	0.91	53
Overall accuracy			0.89	79
Macro average	0.87	0.89	0.87	79
Weighted average	0.89	0.89	0.89	79

5.3 Confusion Matrix

Derived from the reported precision, recall and class sizes, the test-set confusion matrix is:

	Predicted Fail	Predicted Pass
Actual Fail	23	3
Actual Pass	6	47

Of 79 students, 70 were classified correctly and 9 incorrectly. Three failing students were wrongly predicted to pass, and six passing students were wrongly predicted to fail.

5.4 Interpretation

- The model detects **88% of students who will fail** (Fail recall 0.88), which is the most useful property for an early-warning system.
- When it predicts Pass, it is correct 94% of the time.
- Fail precision is lower (0.79) because six passing students were flagged as at risk. For a support-oriented use case, this is an acceptable cost, since extra attention for a student who would have passed anyway is low-risk.

5.5 Sample Prediction

A new student with study time 3, failures 0, absences 5, G1 = 12 and G2 = 13 was passed through the same scaler and model. The model predicted **PASS**, which is consistent with the student's good prior grades and clean failure record.

6. Discussion and Limitations

1. **Dependence on earlier grades.** G1 and G2 are closely related to the final grade G3, so much of the model's accuracy likely comes from these two features. This makes the model strong at predicting the result late in the term, but less helpful for very early prediction before any grades exist. A useful extension is to compare the model with and without G1 and G2.
2. **Small dataset and single split.** With 395 students and only 79 in the test set, each misclassified student changes accuracy by about 1.3 percentage points. Results from one random split may vary; k-fold cross-validation would give a more reliable estimate.

3. **No baseline comparison.** Because about 67% of students pass, a model that always predicts Pass would score roughly 67% accuracy. The model's 88.6% is clearly better, but this baseline should be stated when reporting the result.
4. **Single algorithm.** Only Logistic Regression was tried, with no hyperparameter tuning or comparison against other models.
5. **Limited feature set.** Only five of the 32 available attributes were used. Demographic, family and lifestyle variables were not tested.
6. **Outliers.** Extreme absence values (up to 75) were left untreated. Scaling reduces but does not remove their influence.
7. **Scope.** The data covers one school system and the Mathematics course only, so the model may not generalise to other subjects, schools or countries.
8. **Fairness.** If extended to use personal attributes such as sex, family background or alcohol use, care is needed to avoid unfair or biased predictions about individual students.

7. Future Work

- Apply k-fold cross-validation and report mean and standard deviation of scores.
- Compare Logistic Regression with Decision Tree, Random Forest, SVM and gradient boosting.
- Run an experiment excluding G1 and G2 to build a true early-prediction model.
- Add categorical features using one-hot encoding and evaluate their contribution.
- Examine model coefficients or feature importance to explain which factors drive predictions.
- Treat the target as a regression problem to predict the actual G3 grade.
- Combine the Mathematics and Portuguese datasets for a larger training set.
- Deploy the model as a simple web form for teachers to enter student data and get a prediction.

8. Conclusion

This project developed and evaluated a Logistic Regression model that predicts whether a student will pass or fail the final exam. Trained on 316 students and tested on 79, it achieved 88.6% accuracy with a Pass precision of 0.94, Pass recall of 0.89, and F1-score of 0.91, and it identified 88% of failing students. The results confirm that prior grades, past

failures, absences and study time carry strong predictive signal for exam outcomes. With further validation, broader feature use and model comparison, this approach could become a practical early-warning tool that helps teachers target support to students who need it most.

9. References

1. Cortez, P. and Silva, A. (2008). Using Data Mining to Predict Secondary School Student Performance. In *Proceedings of the 5th Future Business Technology Conference (FUBUTEC 2008)*, pp. 5 to 12.
2. UCI Machine Learning Repository: *Student Performance* dataset.
3. Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825 to 2830.